1. Data Types

1. Find the data type of every value in a mixed list.
2. Convert a nested list into a tuple of tuples.
3. Remove all duplicate values from a mixed list while preserving the original order.

2. Operators

4. Check whether a number is a power of 2 using operators.
5. Swap two numbers using bitwise XOR.
6. Find whether a number is divisible by both 4 and 6 using logical operators.
7. Calculate the total electricity bill using different unit rates.

3. Conditional Statements

8. Check whether three sides can form a triangle.
9. Determine the type of triangle (Equilateral, Isosceles, Scalene).
10. Create a simple ATM menu (Withdraw, Deposit, Balance).
11. Calculate income tax based on different tax slabs.
12. Create a menu-driven calculator using if-elif.

4. Loops

13. Print all prime numbers between 1 and n.
14. Find the factorial of a number using a loop.
15. Print the Fibonacci series up to n terms.
16. Check whether a number is an Armstrong number.
17. Reverse a number and check if it is a palindrome.
18. Find the Greatest Common Divisor (GCD) of two numbers.

5. Functions

19. Write a function to check if a string is a palindrome.
20. Write a function to count vowels and consonants in a string.
21. Create a function to calculate simple and compound interest.
22. Write a function to return all factors of a number.
23. Write a function to find the second-largest number in a list.

6. Lists

24. Merge two lists without duplicates.
25. Find the second-largest and second-smallest elements.

26. Rotate a list to the left by k positions.
27. Separate even and odd numbers into two lists.
28. Find the common elements between two lists.

7. Tuples & Sets

29. Count the frequency of each element in a tuple.
30. Find the union, intersection, and difference of two sets.
31. Check whether one set is a subset of another.

8. Dictionaries

32. Count the frequency of words in a sentence.
33. Create a dictionary from two lists (keys and values).
34. Sort a dictionary by its values.

9. File Handling

35. Read a text file and count the number of lines, words, and characters.
36. Copy only the even-numbered lines from one file to another.

10. Exception Handling

37. Handle invalid integer input using try-except.
38. Handle file-not-found errors while reading a file.

11. Modules

39. Create a random password generator using the random and string modules.
40. Use the datetime module to calculate the number of days between two dates.